Testing Documentation

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R1 was configured with the subnet 192.168.10.0/24. R2 was configured to provide NAT between R1 and R3 and the Internet. R3 was configured with the subnet of 192.168.40.0/24.

To verify connectivity between PC1 and R1, the ping command was issued from interface of Eth0 on PC1 (192.168.10.67) to Eth0/0 on R1 (192.168.10.1). The screen of PC1 below will show that the test was successful.

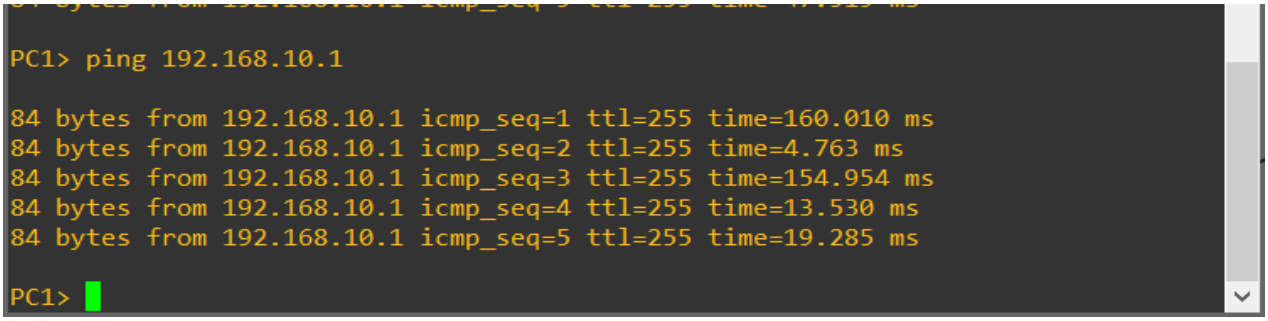


Figure 1 - ping from PC1 to R1

To verify connectivity between PC1 and R2, the ping command was issued from interface of Eth0 on PC1 (192.168.10.67) to Eth0/0 on R2 (192.168.20.2). The screen of PC1 below will show that the test was successful.

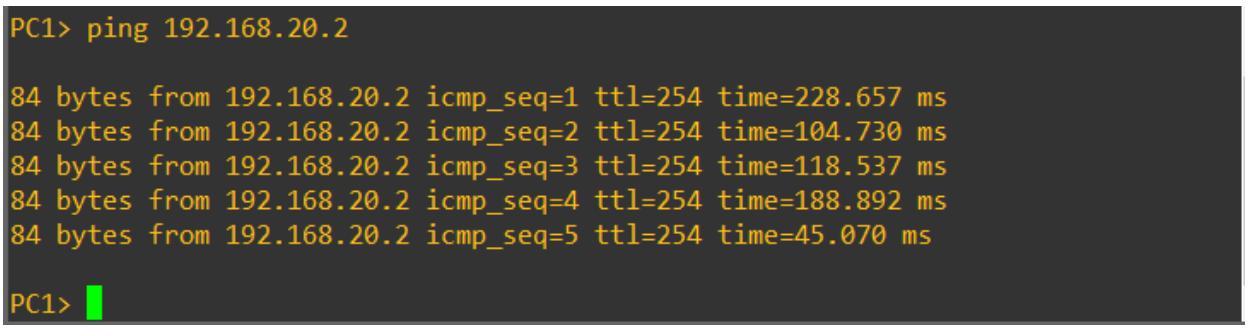


Figure 2 - ping from PC1 to R2

To verify connectivity between PC1 and the Internet, the ping command was issued from interface of Eth0 on PC1 (192.168.10.67) to Eth0/0 on R2 (192.168.83.2). The screen of

PC1 below will show that the test was successful.

```
PC1> ping 192.168.83.2

84 bytes from 192.168.83.2 icmp_seq=1 ttl=126 time=18.803 ms
84 bytes from 192.168.83.2 icmp_seq=2 ttl=126 time=17.854 ms
84 bytes from 192.168.83.2 icmp_seq=3 ttl=126 time=23.362 ms
84 bytes from 192.168.83.2 icmp_seq=4 ttl=126 time=14.997 ms
84 bytes from 192.168.83.2 icmp_seq=5 ttl=126 time=21.090 ms

PC1>
```

Figure 3 - ping from PC1 to Internet

To verify connectivity between PC1 and R3, the ping command was issued from interface of Eth0 on PC1 (192.168.10.67) to Eth0/0 on R3 (192.168.40.1). The screen of PC1 below will show that the test was successful.

```
PC1> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=253 time=2.637 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=253 time=5.926 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=253 time=2.343 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=253 time=2.562 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=253 time=2.309 ms

PC1>
```

Figure 4 - ping from PC1 to R3

To verify connectivity between PC1 and the Ubuntu Server, the ping command was issued from interface of Eth0 on PC1 (192.168.10.67) to Eth0 on Ubuntu Server

(192.168.40.100). The screen of PC1 below will show that the test was successful.

```
PC1> ping 192.168.40.100

84 bytes from 192.168.40.100 icmp_seq=1 ttl=61 time=4.602 ms
84 bytes from 192.168.40.100 icmp_seq=2 ttl=61 time=3.829 ms
84 bytes from 192.168.40.100 icmp_seq=3 ttl=61 time=11.487 ms
84 bytes from 192.168.40.100 icmp_seq=4 ttl=61 time=8.966 ms
84 bytes from 192.168.40.100 icmp_seq=5 ttl=61 time=4.628 ms

PC1>
```

Figure 5 - ping from PC1 to Ubuntu server

To verify connectivity between PC2 and R1, the ping command was issued from interface of Eth0 on PC1 (192.168.10.68) to Eth0/0 on R1 (192.168.10.1). The screen of PC2 below will show that the test was successful.

```
PC2> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.590 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.994 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.312 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.408 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.060 ms

PC2>
```

Figure 6 - ping from PC2 to R1

To verify connectivity between PC2 and R2, the ping command was issued from interface of Eth0 on PC1 (192.168.10.68) to Eth0/0 on R2 (192.168.20.2). The screen of PC2 below will show that the test was successful.

```
PC2> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.783 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.306 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.599 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.597 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.148 ms

PC2>
```

Figure 7 - ping from PC2 to R2

To verify connectivity between PC2 and the Internet, the ping command was issued from interface of Eth0 on PC1 (192.168.10.68) to Eth0/0 on R2 (192.168.83.2). The screen of PC2 below will show that the test was successful.

```
PC2> ping 192.168.83.2

84 bytes from 192.168.83.2 icmp_seq=1 ttl=126 time=14.907 ms
84 bytes from 192.168.83.2 icmp_seq=2 ttl=126 time=30.620 ms
84 bytes from 192.168.83.2 icmp_seq=3 ttl=126 time=28.855 ms
84 bytes from 192.168.83.2 icmp_seq=4 ttl=126 time=16.353 ms
84 bytes from 192.168.83.2 icmp_seq=5 ttl=126 time=14.218 ms

PC2>
```

Figure 8 - ping from PC2 to Internet

To verify connectivity between PC2 and R3, the ping command was issued from interface of Eth0 on PC1 (192.168.10.68) to Eth0/0 on R3 (192.168.40.1). The screen of PC2 below will show that the test was successful.

```
PC2> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=253 time=6.775 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=253 time=5.362 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=253 time=2.684 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=253 time=2.620 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=253 time=3.242 ms

PC2>
```

Figure 9 - ping from PC2 to R3

To verify connectivity between PC2 and the Ubuntu Server, the ping command was issued from interface of Eth0 on PC1 (192.168.10.68) to Eth0 on Ubuntu Server

(192.168.40.100). The screen of PC2 below will show that the test was successful.

```
PC2> ping 192.168.40.100

84 bytes from 192.168.40.100 icmp_seq=1 ttl=61 time=8.354 ms
84 bytes from 192.168.40.100 icmp_seq=2 ttl=61 time=5.488 ms
84 bytes from 192.168.40.100 icmp_seq=3 ttl=61 time=12.549 ms
84 bytes from 192.168.40.100 icmp_seq=4 ttl=61 time=35.048 ms
84 bytes from 192.168.40.100 icmp_seq=5 ttl=61 time=13.748 ms

PC2>
```

Figure 10 - ping from PC2 to Ubuntu server

To verify connectivity between Kali Linux and R1, the ping command was issued from interface of Eth0 on PC1 (192.168.10.69) to Eth0/0 on R1 (192.168.10.1). The screen of Kali Linux below will show that the test was successful.

```
File  Actions  Edit  View  Help
┌──(lsidner㉿kali)-[~]
└─$ ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=5.64 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=3.16 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=1.29 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=4.79 ms
64 bytes from 192.168.10.1: icmp_seq=5 ttl=255 time=2.26 ms
64 bytes from 192.168.10.1: icmp_seq=6 ttl=255 time=3.92 ms
^C
--- 192.168.10.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5012ms
rtt min/avg/max/mdev = 1.293/3.511/5.640/1.468 ms

┌──(lsidner㉿kali)-[~]
└─$
```

Figure 11 - ping from Kali Linux to R1

To verify connectivity between Kali Linux and R2, the ping command was issued from interface of Eth0 on PC1 (192.168.10.69) to Eth0/0 on R2 (192.168.20.2). The screen of Kali

Linux below will show that the test was successful.

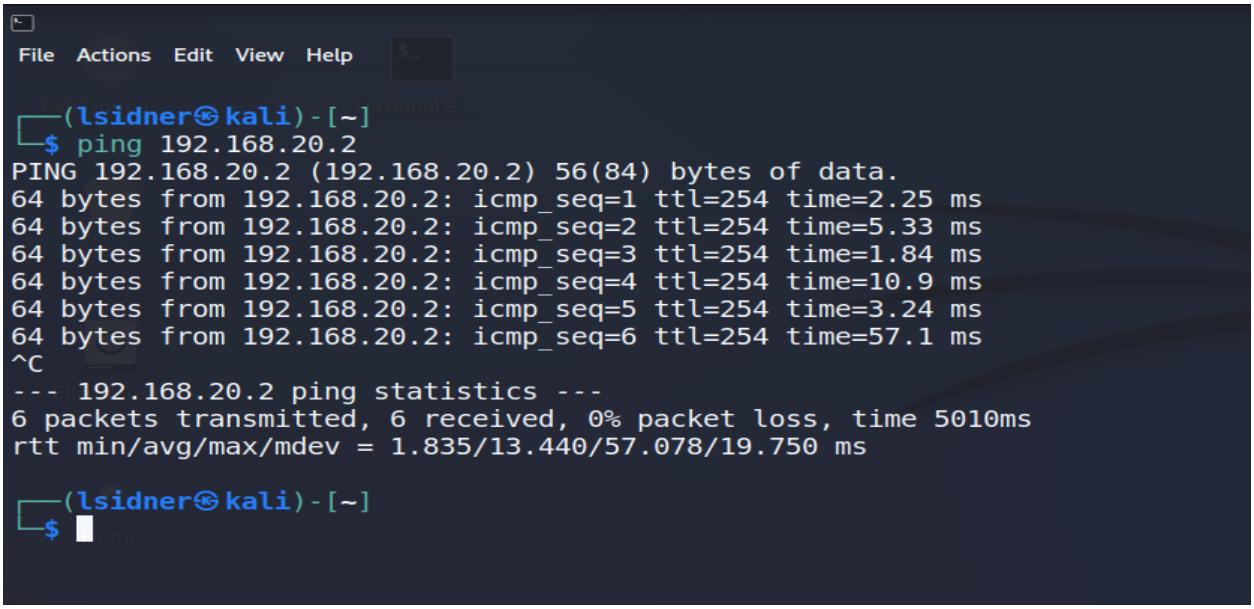


Figure 12 - ping from Kali Linux to R2

To verify connectivity between Kali Linux and the Internet, the ping command was issued from interface of Eth0 on PC1 (192.168.10.69) to Eth0/0 on R2 (192.168.83.2). The screen of Kali Linux below will show that the test was successful.

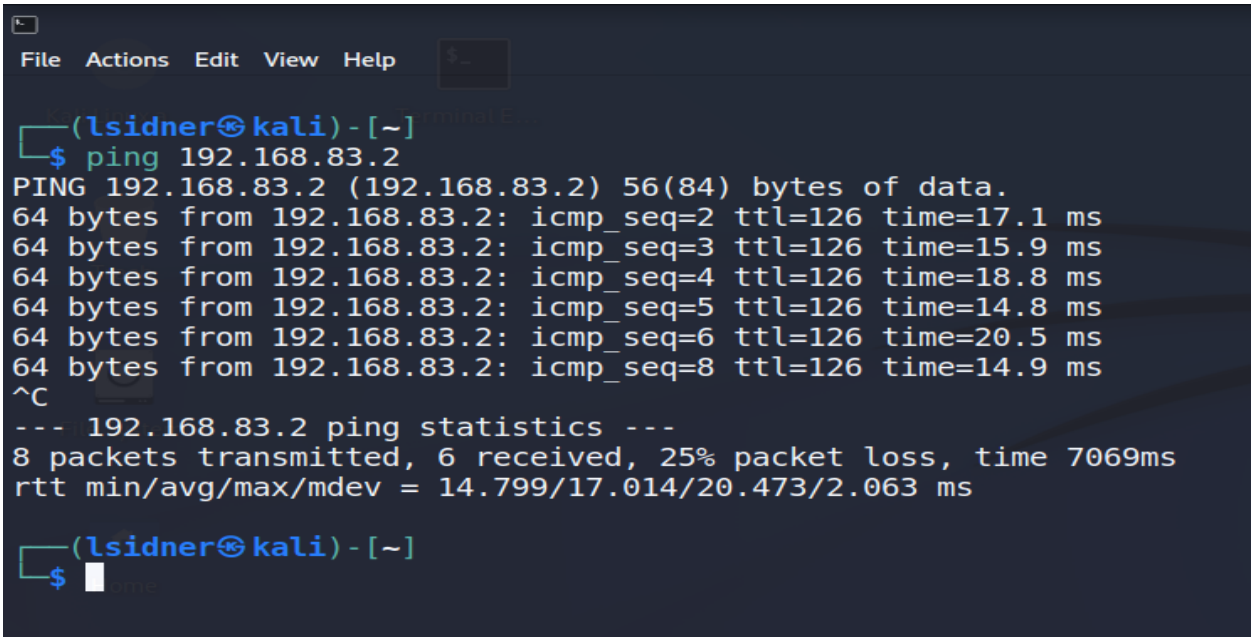


Figure 13 - ping from Kali Linux to Internet

To verify connectivity between Kali Linux and R3, the ping command was issued from interface of Eth0 on PC1 (192.168.10.69) to Eth0/0 on R3 (192.168.40.1). The screen of Kali Linux below will show that the test was successful.

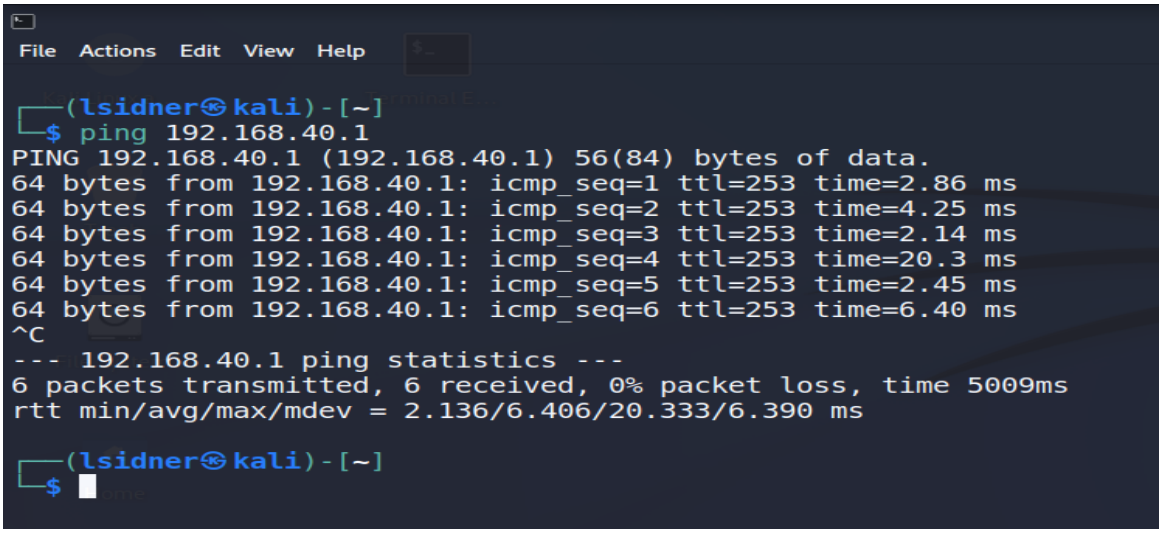


Figure 14 - ping from Kali Linux to R3

To verify connectivity between Kali Linux and the Ubuntu Server, the ping command was issued from interface of Eth0 on PC1 (192.168.10.69) to Eth0 on Ubuntu Server (192.168.40.100). The screen of Kali Linux below will show that the test was successful.

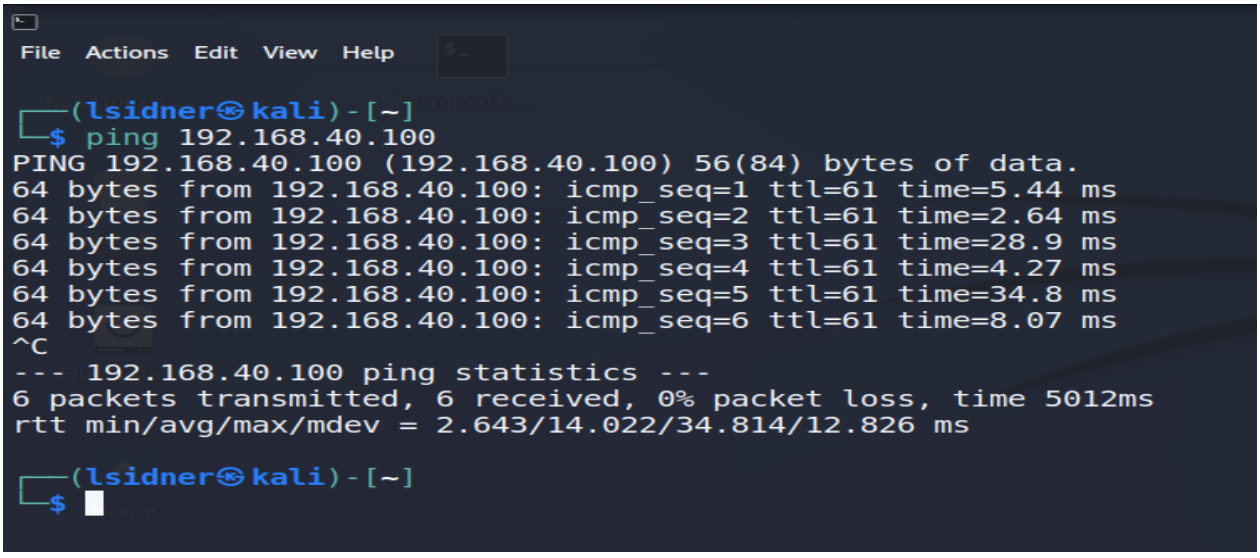


Figure 15 - ping from Kali Linux to Ubuntu server

To verify connectivity between R1 and R3, the ping command was issued from interface Eth0/1 on R1 (192.168.10.1) to Eth0/0 on R3 (192.168.40.1). The screen of R1 below will show that the test was successful.

```
R1#ping 192.168.40.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/5/7 ms
R1#
```

Figure 16 - ping from R1 to R3

To verify the routing table of R1, the sh ip route command was entered. Below is the screen of R1 to show the routing table. The following routes are shown: local, connected, static, OSPF, and OSPF inter area. The connected routes are the subnet of 192.168.10.0/24, 192.168.10.64/29, and 192.168.20.0/24. The local routes are the actual address of the interface where E0/0 is 192.168.10.1, E0/0.10 is 192.168.10.65, and E0/1 is 192.168.20.1. The OSPF route is area 0 which includes the subnet between R2 and R3, which is 192.168.30.0/24. The OSPF inter area route is area 2 which is the subnet of R3, which is 192.168.40.0/24. The static route is

the default route to the internet.

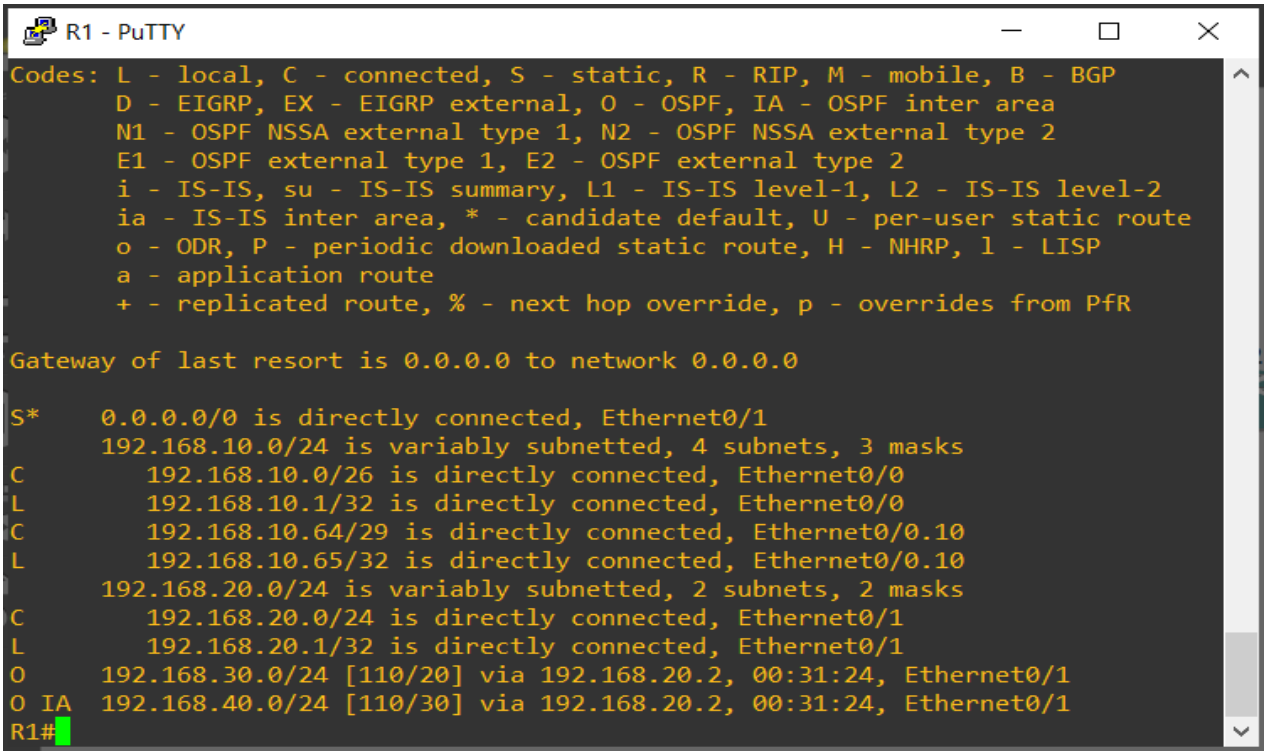


Figure 17 - R1 routing table

To verify the routing table of R2, the sh ip route command was entered. Below is the screen of R2 to show the routing table. The following routes are shown: local, connected, static, OSPF, and OSPF inter area. The connected routes are the subnet of 192.168.20.0/24, 192.168.30.0/24, and 192.168.83.0/24. The local routes are the actual address of the interface where E0/0 is 192.168.20.2 and E0/1 is 192.168.30.1. The OSPF inter area routes are the areas 1 and 2. Area 1 includes the subnets of R1, which are 192.168.10.0/24 and 192.168.10.64/29. Area 2 contains the subnet of R3 which is 192.168.40.0/24. The static route is the default route

to the internet.

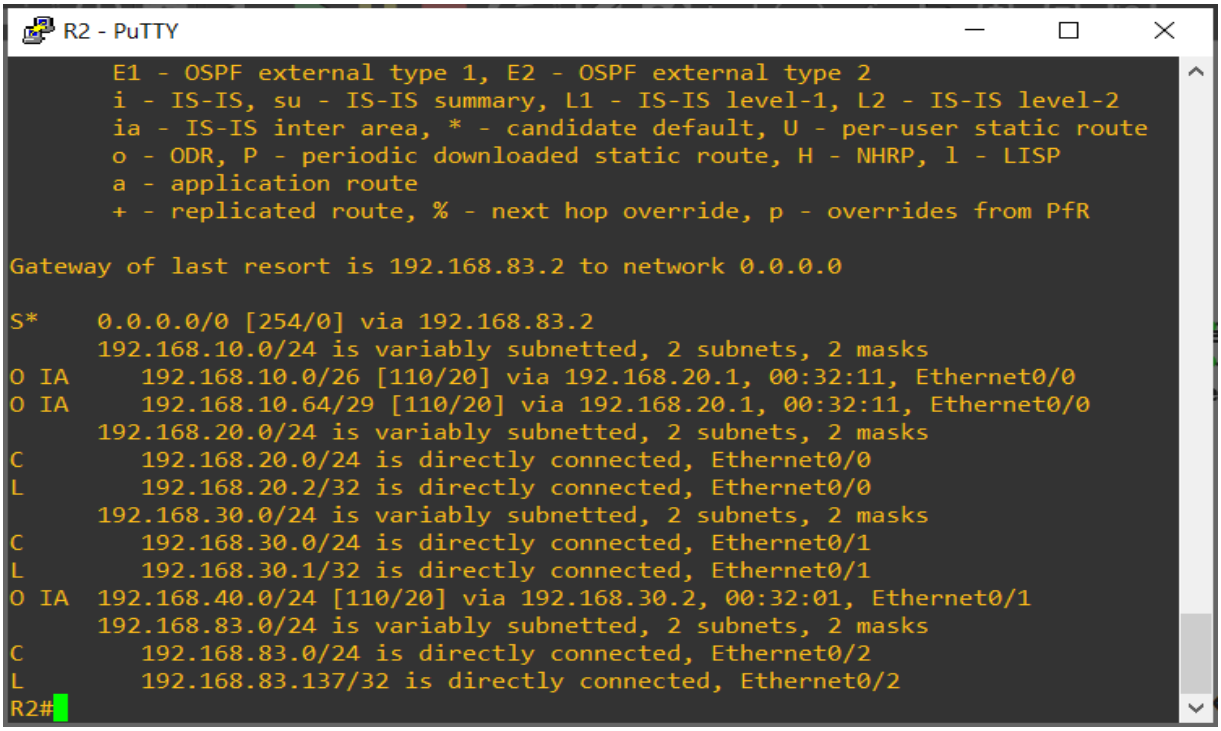


Figure 18 - R2 routing table

To verify the routing table of R3, the sh ip route command was entered. Below is the screen of R3 to show the routing table. The following routes are shown: local, connected, static, OSPF, and OSPF inter area. The connected routes are the subnet of 192.168.40.0/24 and 192.168.30.0/24. The local routes are the actual address of the interface where E0/0 is 192.168.30.2 and E0/1 is 192.168.40.1. The OSPF route is the area 0 subnet which is 192.168.20.0/24 which is the subnet of R2. The OSPF inter area routes are a part of area 1. Area 1 area includes the subnets of R1 which are 192.168.10.0/24 and 192.168.10.64/29. The static

route is the default route to the internet.

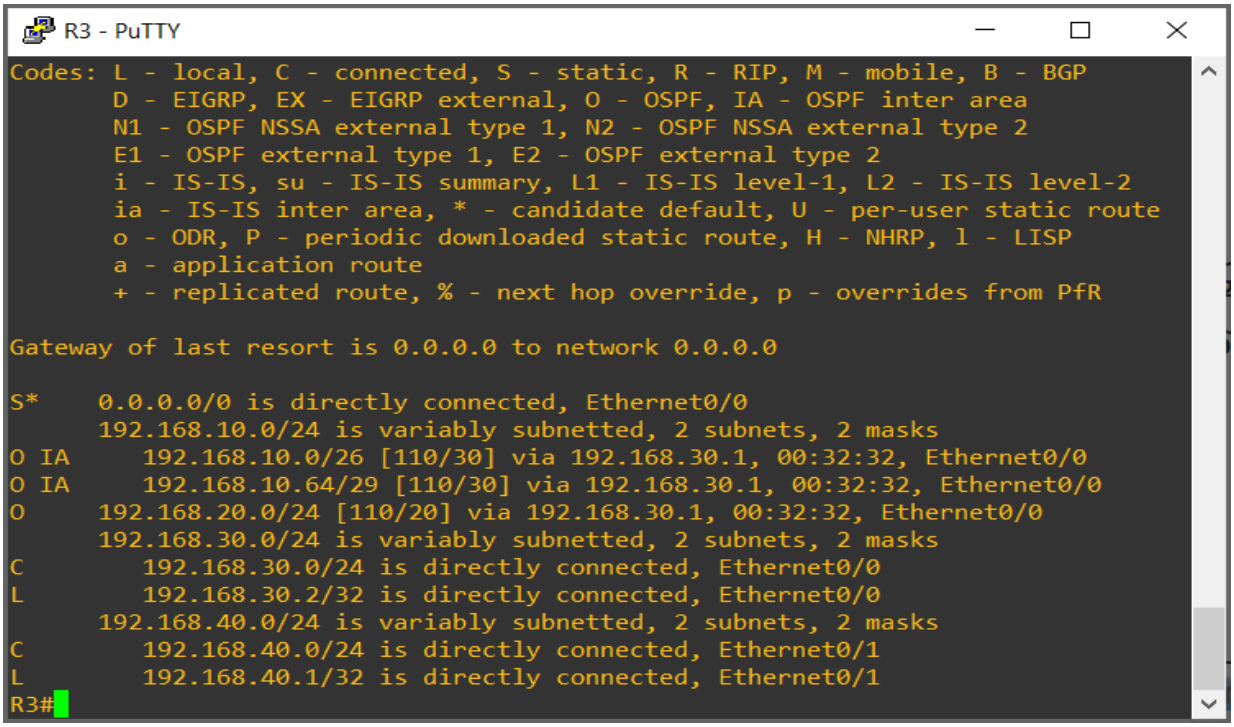


Figure 19 - R3 routing table

To verify the access lists were created, the sh ip access-lists was entered. The extended access-list (ACL) 100 was configured to include subnets of R1, R2, and R3. These subnets were able to connect to the internet via 192.168.83.2. The ACL was tested using the ping command from PC1 which was able to test connectivity to the internet. Below is the screen of R2 that shows the created ACL. The 10 matches next to the first line, permit ip 192.168.10.0 0.0.0.255 host 192.168.83.2, indicate that there was 10 times that the ACL found a match with the subnet of R1.

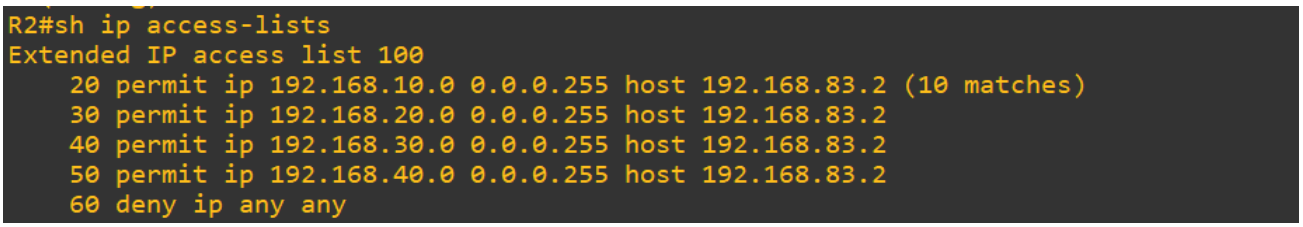


Figure 20 - access-list 100 on R2